

企業資料通訊期末專題

Please use MS WORD to do your homework and submit your document to i-learning

請同學直接將圖片填貼在附檔 word 檔案中，上傳至 i-learning 平台。

需求描述：資管樓網路布置方案

背景：

資管樓計劃今天進行網路布置，需要為一樓、二樓和三樓配置網路設備及與其相應的網段。每層樓將擁有獨立的網段，並且各層樓之間通過專線進行互聯。

詳細需求：

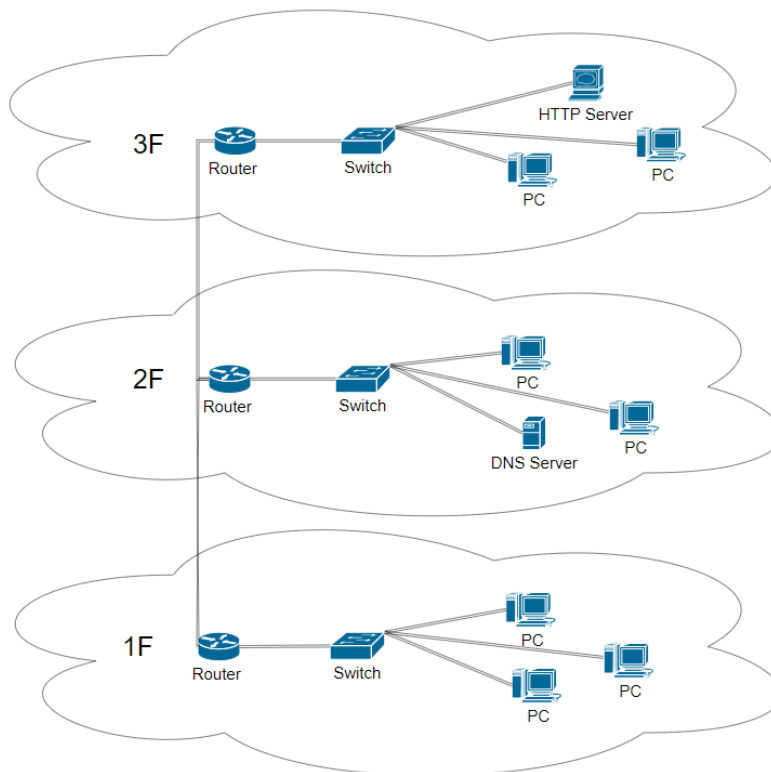
一樓配置：1. 三台電腦 2. 獨立的網段 3. 防火牆(DNS使用TCP和UDP協議，端口是53)

二樓配置：1. 兩台電腦和一台DNS伺服器 2. 獨立的網段

三樓配置：1. 兩台電腦和一台HTTP伺服器 2. 獨立的網段

專線連接：

1. 各層樓之間的路由器需要通過專線進行連接，以實現不同網段之間的通信。
2. 以二樓的Router作為核心連接點，負責與一樓和三樓的路由器相互連接，並管理各層樓間數據流的轉發。



上圖為資管樓網路架構示意圖

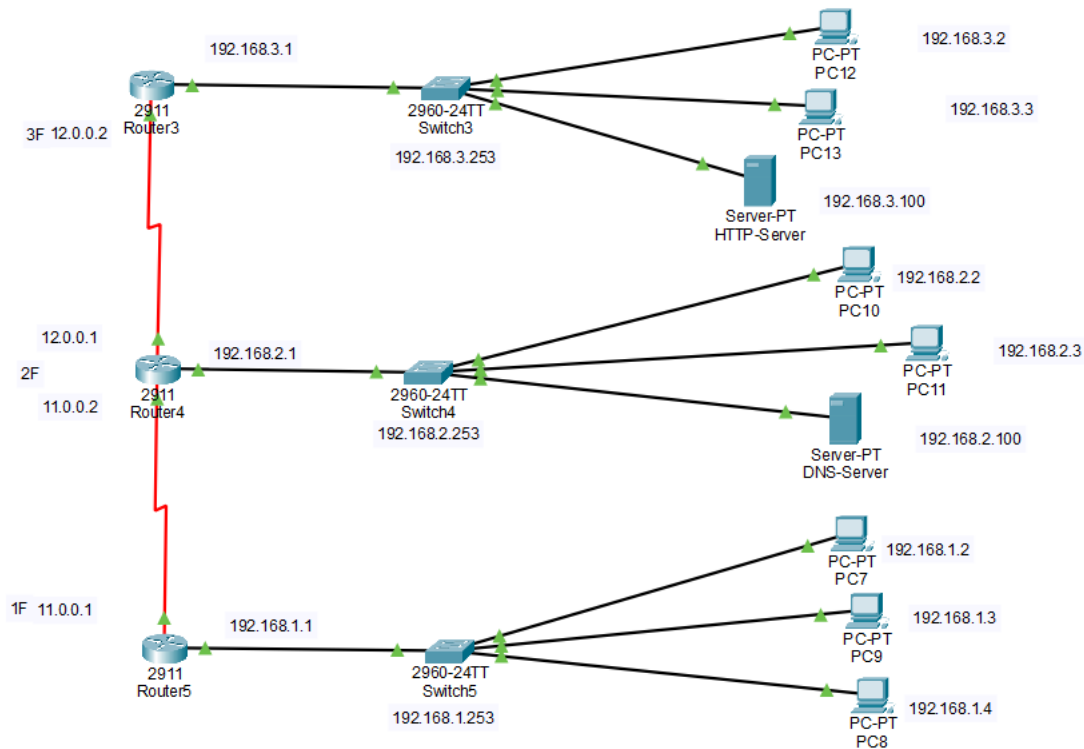
1. 請同學將設定完成的設備IP位置與子網路遮罩填入下方的表格中。

答案：(請將答案填寫在此處)

樓層	設備	IP	子網路遮罩
Ex. 1F	PC	192.168.1.99	255.255.255.0
1F	Router	192.168.1.1	255.255.255.0
1F	Switch	192.168.1.253	255.255.255.0
1F	PC	192.168.1.2	255.255.255.0
1F	PC	192.168.1.3	255.255.255.0
1F	PC	192.168.1.4	255.255.255.0
2F	Router	192.168.2.1	255.255.255.0
2F	Switch	192.168.2.253	255.255.255.0
2F	PC	192.168.2.2	255.255.255.0
2F	PC	192.168.2.3	255.255.255.0
2F	DNS Server	192.168.2.100	255.255.255.0
3F	Router	192.168.3.1	255.255.255.0
3F	Switch	192.168.3.253	255.255.255.0
3F	PC	192.168.3.2	255.255.255.0
3F	PC	192.168.3.3	255.255.255.0
3F	HTTP Server	192.168.3.100	255.255.255.0

2. 請同學將完整的網路截圖貼上WORD檔。

答案：(請將答案填寫在此處)



3. 請同學將二樓Router的路由表截圖貼上WORD檔。

答案：(請將答案填寫在此處)

```
Router>show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

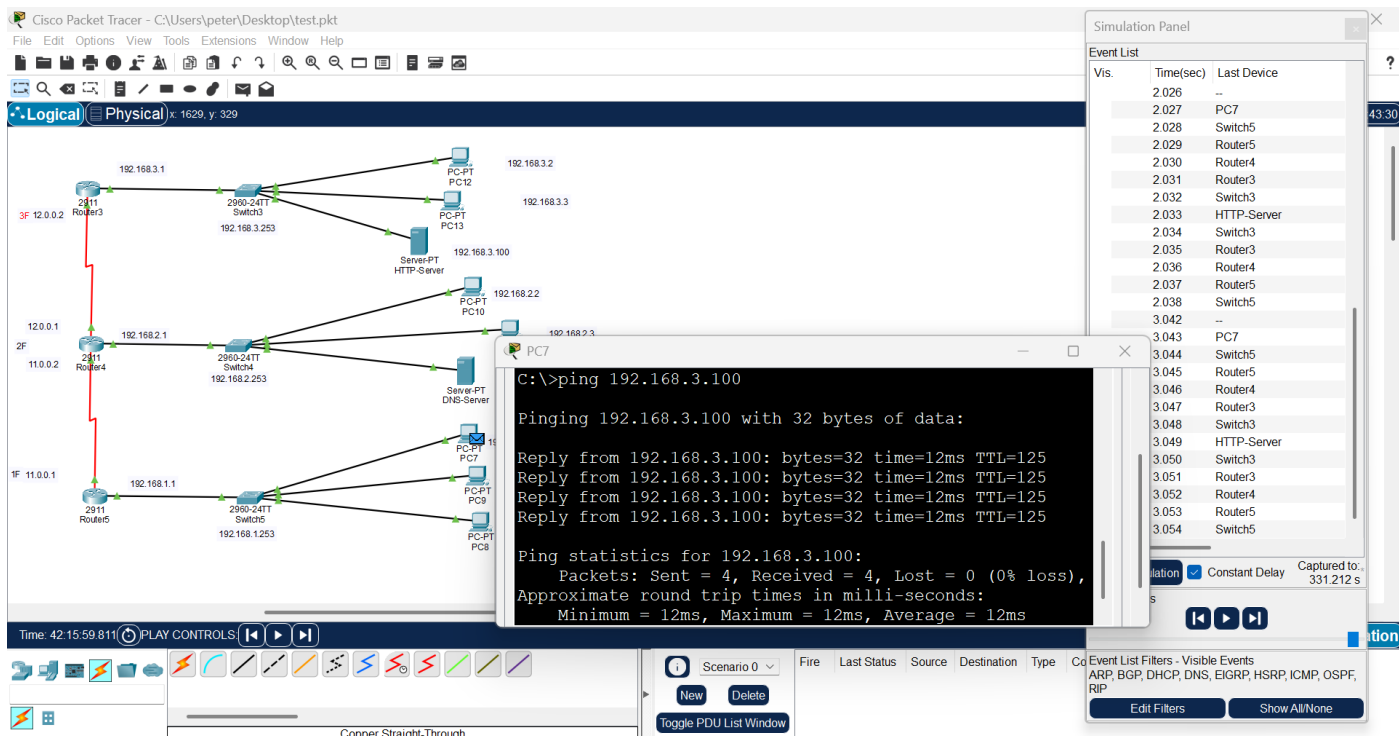
```
11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    11.0.0.0/8 is directly connected, Serial0/0/0
L    11.0.0.2/32 is directly connected, Serial0/0/0
12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Serial0/0/1
L    12.0.0.1/32 is directly connected, Serial0/0/1
S    192.168.1.0/24 [1/0] via 11.0.0.1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/0
L    192.168.2.1/32 is directly connected, GigabitEthernet0/0
S    192.168.3.0/24 [1/0] via 12.0.0.2
```

```
Router>
```

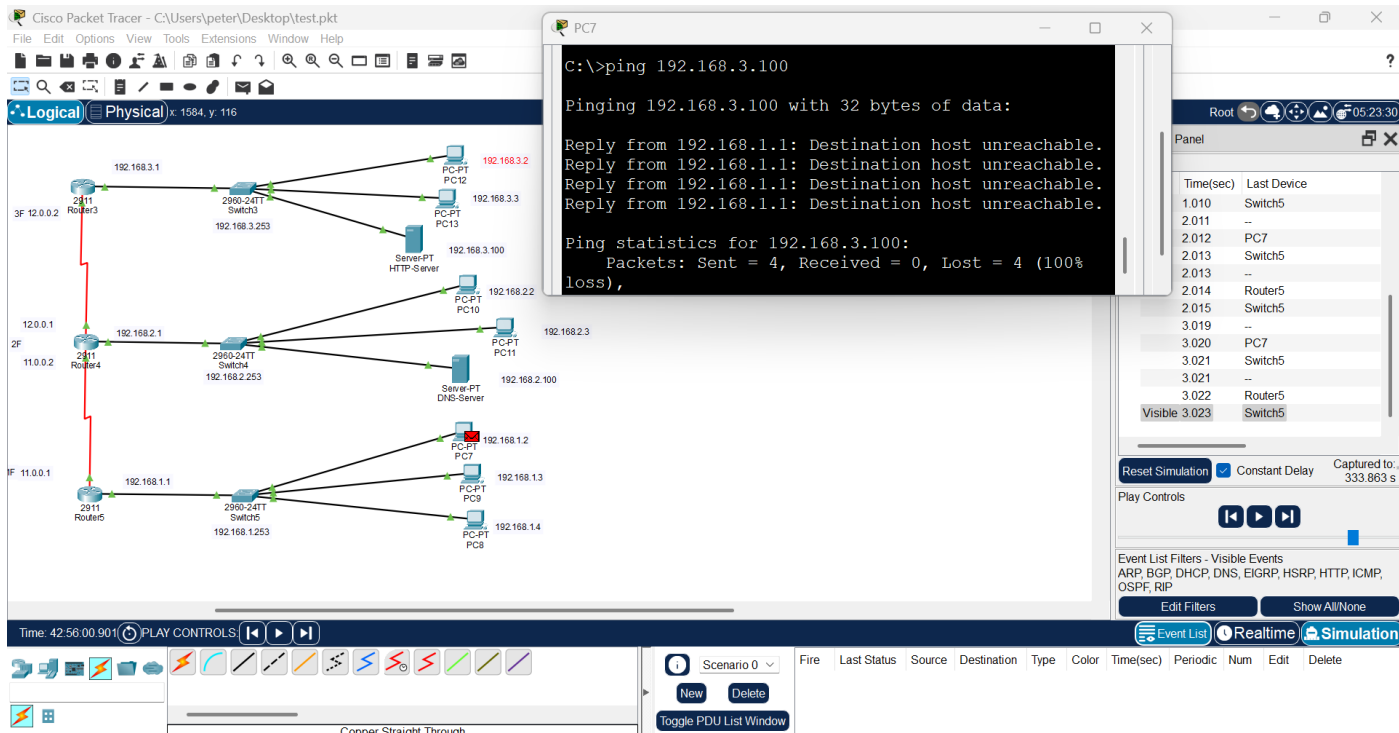
4. 請同學將「從一樓電腦Ping HTTP Server的畫面和封包繞送路線」截圖並貼上WORD檔。

答案：(請將答案填寫在此處)

1F ROUTER 無防火牆：



1F ROUTER 有防火牆：



5. 如果今天老師在一樓的電腦上瀏覽網頁。

請同學將完整的網路圖和瀏覽網頁的封包繞送路線一起截圖並貼上WORD檔。

答案：(請將答案填寫在此處)

Cisco Packet Tracer - C:\Users\peter\Desktop\test.pkt
File Edit Options View Tools Extensions Window Help
[Icons]

Logical Physical x: 1443, y: 285

3F 12.0.0.2 2911 Router3 192.168.3.1 2960-24TT Switch3 192.168.3.253 PC-PT PC12 192.168.3.2 PC-PT PC13 192.168.3.3 Server-PT HTTP-Server 192.168.3.100 PC-PT PC10 192.168.2.2 PC-PT PC11 192.168.2.3 Server-PT DNS-Server 192.168.2.100

2F 12.0.0.1 2911 Router4 192.168.2.1 2960-24TT Switch4 192.168.2.253 PC-PT PC7 192.168.1.2 PC-PT PC9 192.168.1.3 PC-PT PC8 192.168.1.4

1F 11.0.0.1 2911 Router5 192.168.1.1 2960-24TT Switch5 192.168.1.253

PC8

Physical Config Desktop Programming Attributes

Web Browser

< > URL: 1142232.cycu.edu.tw Go Stop

11142232

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:
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[Image](#)

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC8
	0.002	Switch5
	0.003	Router5
	0.004	Router4
	0.005	Switch4
	0.005	Switch4
	0.005	Switch4
	0.006	DNS-Server
	0.007	Switch4
	0.008	Router4
	0.009	Router5
	0.010	Switch5
	0.022	--
	0.023	--
	0.024	PC8
	0.025	Switch5
	0.026	Router5
	0.027	Router4
	0.028	Router3
	0.029	Switch3
	0.030	HTTP-Server
	0.031	Switch3
	0.032	Router3
	0.033	Router4
	0.034	Router5
Visible	0.035	Switch5

Reset Simulation Constant Delay Captured to: 329.875 s

Play Controls

Event List Filters - Visible Events

Time: 42:32:54.018 PLAY CONTROLS [Icons]

[Icons]

Copper Straight-Through

Scenario 0 [New] [Delete] [Toggle PDU List Window]

Fire Last Status Source Destination Type Co